

# MINI WEB APPLICATION – CRUD-BASED WEB APPLICATION

## Student Portfolio Management System

### Submitted By

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## 1. Introduction

The **Student Portfolio Management System** is a web-based application designed to help students create, manage, update and present their academic and technical information in one place.

The application provides a login and registration system and allows each student to maintain their own portfolio. Students can manage their **profile, education, skills, projects and certifications**. The application also provides a separate portfolio view that can be used to present the student's information in an organized manner.

The application was developed as a simple and user-friendly mini web application using **HTML, CSS and JavaScript**. Data is stored locally using **Browser LocalStorage**, allowing the information to remain available even after refreshing or logging out and logging in again. The application implements CRUD operations for portfolio information.

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## 2. Objective

The main objectives of this project are:

- To develop a functional mini web application.

- To understand frontend web application development.
  - To implement **CRUD operations**.
  - To provide a simple interface for managing student information.
  - To store and retrieve user data.
  - To implement editing and deletion of existing records.
  - To create a presentable student portfolio.
  - To understand data persistence using LocalStorage.
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## 3. Technologies Used

| Technology   | Purpose                                        |
|--------------|------------------------------------------------|
| HTML5        | Structure of the web application               |
| CSS3         | Styling and responsive user interface          |
| JavaScript   | Application logic and CRUD operations          |
| LocalStorage | User authentication and portfolio data storage |
| Browser      | Running and testing the application            |

The application uses LocalStorage to maintain user accounts and individual portfolio data. Portfolio information is stored using a user-specific key.

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## 4. Features of the Application

The major features of the Student Portfolio Management System are:

### 4.1 User Registration

New users can create an account by entering their name, username and password. The application validates the input before creating the account.

### 4.2 User Login

Registered users can log in using their username and password. The application verifies the stored credentials before opening the portfolio dashboard.

### 4.3 Profile Management

Students can maintain basic profile information such as their name, role and biography.

### 4.4 Education Management

Students can add their educational qualifications and can later **view, edit, update and remove** them.

### 4.5 Skills Management

Students can add technical or soft skills. Existing skills can be edited or removed.

### 4.6 Project Management

Students can maintain their project information, including:

- Project title
- Technologies used
- Description
- Project link

Projects support adding, viewing, editing, updating and deleting.

### 4.7 Certification Management

Students can add certificates, including the certificate title, issuing organization and date. Certificates can also be edited and deleted.

### 4.8 Portfolio View

The application provides a “**View my portfolio**” option that displays the student's information in a separate portfolio presentation.

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## 5. CRUD Operations

CRUD stands for:

- **C – Create**
- **R – Read**
- **U – Update**

- **D – Delete**

The application demonstrates all four operations.

## **Create**

The user enters information through a form and saves it. The new record is added to the portfolio.

## **Read**

Saved records are displayed on the corresponding section of the dashboard.

## **Update**

The **Edit** button loads an existing record into the form. The user can modify the information and save the changes.

The application maintains an editing state and updates the existing record instead of creating a duplicate.

## **Delete**

The user can remove an existing record using the **Remove/Delete** option. A confirmation dialog is displayed before deletion.

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# **6. Data Storage**

The application uses **Browser LocalStorage** instead of an external database.

Each user's portfolio is stored separately using a user-specific storage key. This prevents portfolio information from different demo accounts from being mixed.

The application stores portfolio information using a structure containing:

- Name
- Bio
- Role
- Education
- Skills
- Projects
- Certifications

LocalStorage also allows the information to remain available after refreshing the browser.

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## 7. User Interface

The application uses a simple dashboard layout consisting of:

- Top navigation bar
- Sidebar navigation
- Main content area
- Forms for entering information
- Cards/entries for saved information
- Edit and Delete controls

The sidebar provides access to:

**Profile | Education | Skills | Projects | Certifications**

The application also provides a button to view the completed portfolio and a logout option.

The interface was improved to provide clearer CRUD controls, custom confirmation messages and better spacing while maintaining the original visual design.

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## 8. Validation and Error Handling

The application performs basic validation before saving information.

For example:

- Required fields must be filled.
- Passwords must match during registration.
- Duplicate usernames are rejected.
- Invalid project links are rejected.
- Users receive confirmation messages after successful operations.

The application also provides toast messages for successful or unsuccessful operations.

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## 9. Testing

The application was tested using the following sequence:

| Test                    | Result |
|-------------------------|--------|
| User Registration       | Passed |
| User Login              | Passed |
| Add Profile Information | Passed |
| Add Education           | Passed |
| Edit Education          | Passed |
| Delete Education        | Passed |
| Add Skill               | Passed |
| Edit Skill              | Passed |
| Delete Skill            | Passed |
| Add Project             | Passed |
| Edit Project            | Passed |
| Delete Project          | Passed |
| Add Certification       | Passed |
| Edit Certification      | Passed |
| Delete Certification    | Passed |
| View Portfolio          | Passed |
| Browser Refresh         | Passed |
| Logout and Login Again  | Passed |
| Data Persistence        | Passed |

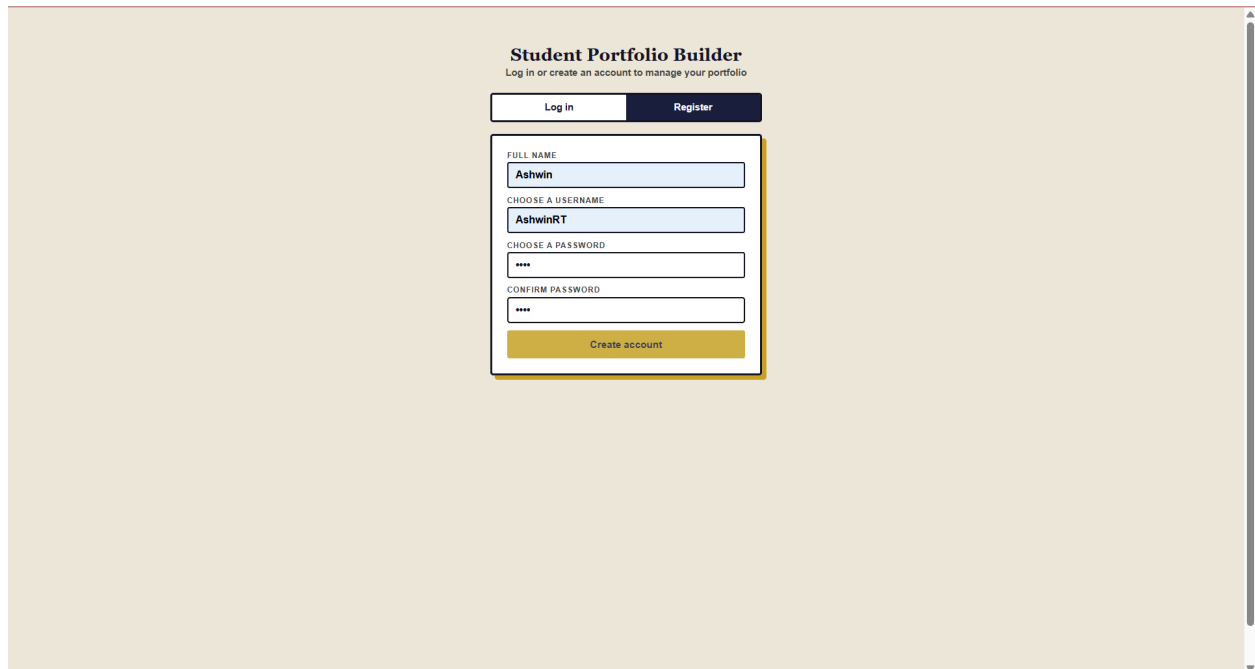
The application's portfolio data is saved using LocalStorage, and the data remains available after refreshing and logging in again.

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# 10. Screenshots

Add your screenshots under this section.

## Screenshot 1 – Login / Registration Page



The screenshot displays the 'Student Portfolio Builder' login and registration interface. At the top, the title 'Student Portfolio Builder' is followed by the subtitle 'Log in or create an account to manage your portfolio'. Below this, there are two buttons: 'Log in' and 'Register'. The 'Register' button is highlighted in dark blue. The registration form is enclosed in a white box with a yellow border. It contains the following fields: 'FULL NAME' with the value 'Ashwin', 'CHOOSE A USERNAME' with the value 'AshwinRT', 'CHOOSE A PASSWORD' with masked characters '\*\*\*\*', and 'CONFIRM PASSWORD' with masked characters '\*\*\*\*'. A yellow 'Create account' button is positioned at the bottom of the form.

**Student Portfolio Builder**  
Log in or create an account to manage your portfolio

Log in Register

FULL NAME  
Ashwin

CHOOSE A USERNAME  
AshwinRT

CHOOSE A PASSWORD  
\*\*\*\*

CONFIRM PASSWORD  
\*\*\*\*

Create account

## Screenshot 2 – Profile / Dashboard

P

Ashwin

View my portfolio

Log out

Profile

Education

Skills

Projects

Certifications

Welcome, Ashwin

Keep your portfolio current and ready to share.

1

EDUCATION

1

SKILLS

1

PROJECTS

1

CERTIFICATES

80% complete

Profile

This is the intro of your portfolio.

FULL NAME

Ashwin

ROLE / TAGLINE (E.G. FIRST-YEAR IT STUDENT)

first-year IT Student

SHORT BIO

Save profile

## Screenshot 3 – Education / Skills Management

P

Ashwin

View my portfolio

Log out

Profile

Education

Skills

Projects

Certifications

Education

Add your college, degree, and year.

Nothing added yet.

DEGREE / COURSE

IT

YEAR

2026-2029

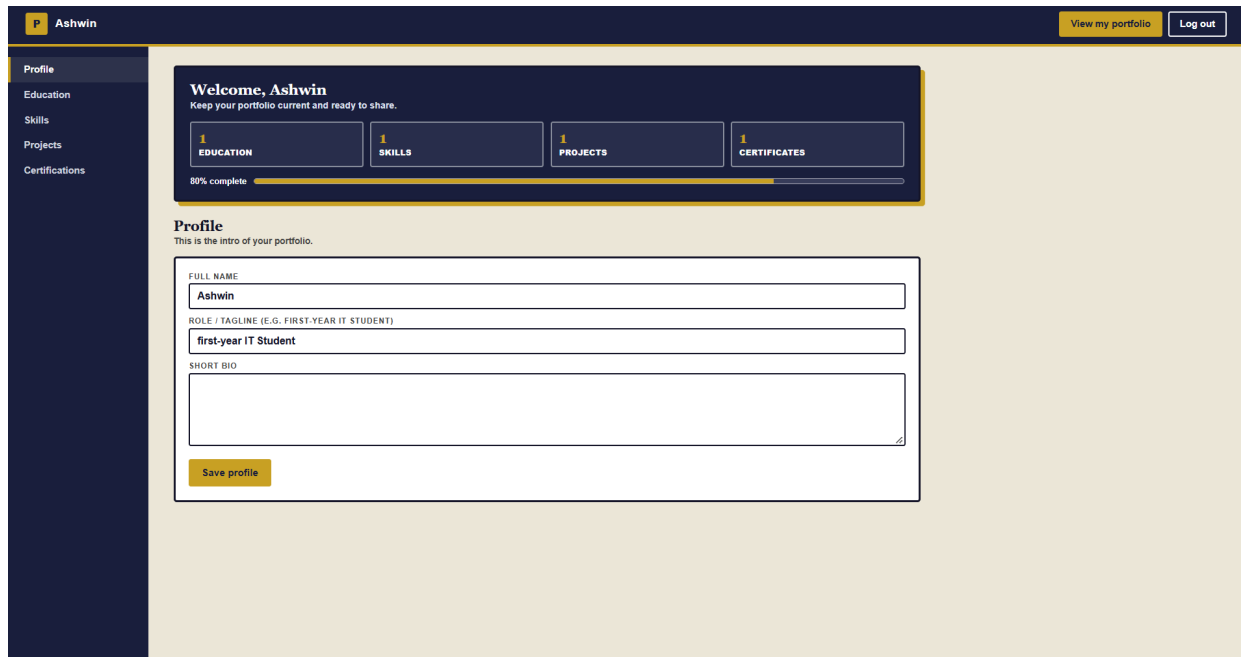
COLLEGE / INSTITUTION

VSBECE

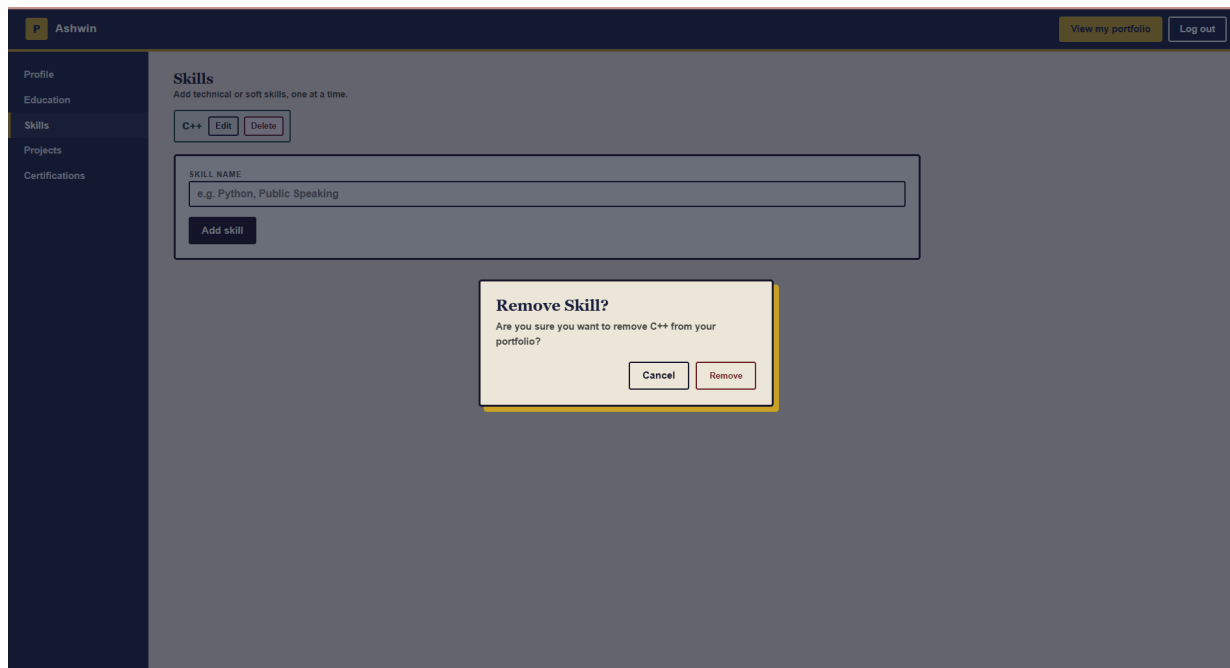
Add education

## Screenshot 4 – Edit / Update Operation





## Screenshot 6 – Delete Confirmation



## Screenshot 7 – Certifications Management

P

Ashwin

View my portfolio

Log out

Profile
Education
Skills
Projects
Certifications

### Certifications

Courses, certificates, or workshops you've completed.

Programming in Java

NPTEL · OCT-2025

Edit

Delete

CERTIFICATE TITLE

e.g. Python for Everybody

DATE

March 2026

ISSUED BY

e.g. Coursera, NPTEL

Add certification

## Screenshot 8 – Final Portfolio View

Back to editing

Ashwin

first-year IT Student

EDUCATION

IT

VSB Engineering College Karur · 2025-2029

SKILLS

C++

PROJECTS

Student Portfolio

HTML, CSS, JS

Student can create a their own portfolio.

CERTIFICATIONS

Programming in Java

NPTEL · OCT-2025

# 11. Advantages

- Simple and easy-to-use interface.

- Provides complete CRUD functionality.
  - Student information can be organized in one place.
  - Data remains available after browser refresh.
  - No additional database installation is required.
  - Portfolio can be viewed separately.
  - Supports multiple portfolio sections.
  - Easy to demonstrate and understand.
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## 12. Limitations

- Data is stored only in the browser using LocalStorage.
  - The application does not use a cloud database.
  - Data is not automatically synchronized between different devices.
  - The authentication system is intended for a college demonstration and is not suitable for production-level security.
  - Clearing browser storage will remove locally stored application data.
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## 13. Future Enhancements

The application can be improved in the future by adding:

- Firebase or another cloud database.
  - Secure server-side authentication.
  - Profile photo upload.
  - Resume upload/download.
  - Portfolio sharing through a public URL.
  - Custom portfolio themes.
  - Deployment to a web hosting service.
  - Admin or institutional features.
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## 14. Result

The **Student Portfolio Management System** was successfully developed as a functional CRUD-based web application.

The application allows users to register and log in, manage their profile, education, skills, projects and certifications, and view their information as a portfolio. CRUD operations were implemented successfully, and LocalStorage was used to maintain the user's data between sessions.

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## **15. Conclusion**

The project provided practical experience in developing a web application using HTML, CSS and JavaScript. It helped in understanding important concepts such as user interaction, form handling, data storage, CRUD operations, validation and responsive interface design.

The completed application provides a simple way for students to maintain and present their academic and technical achievements through a personal portfolio.